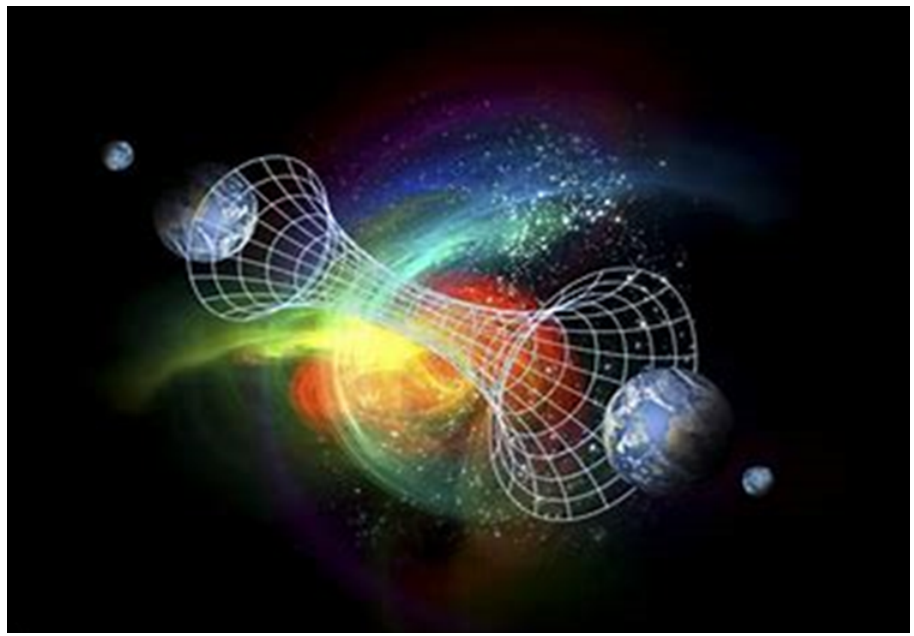
Difference-in-Differences

The Canonical Design: Comparison Groups, Dynamics, Inference

F. Candau (UPPA-TREE)

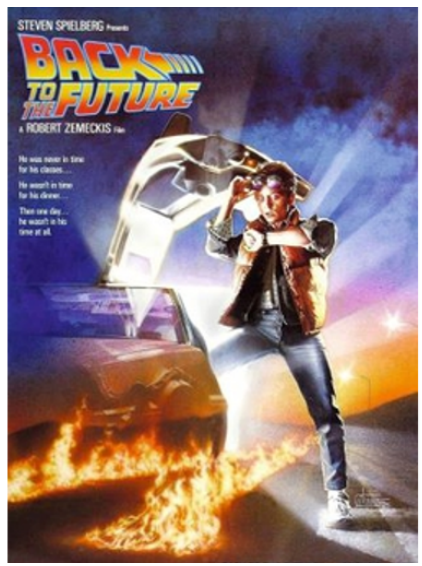


Causal inference and the counterfactual

Difference-in-differences (hereafter **DiD**) is a method of **causal inference** — it identifies a cause-and-effect relationship. It turns on one question: *what would have happened had the shock or policy never occurred?* From that **counterfactual** we recover the effect of the shock or the policy.

Science fiction is built on the same question — stories of an alternative history or a parallel world: *The Man in the High Castle* (P. K. Dick), *Back to the Future*, *Doctor Strange* (Marvel), and many more.







What is difference-in-differences?

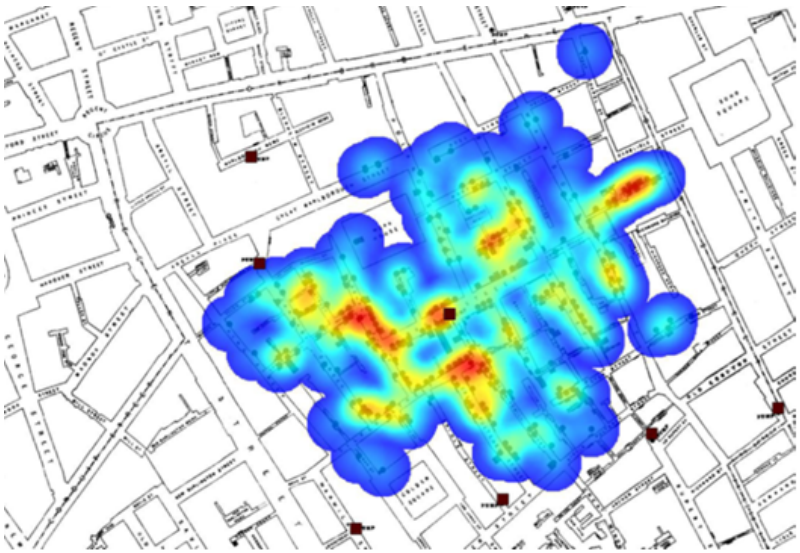
An exogenous shock (a reform, a natural disaster, a price change) hits one group of units, the **treated group**, at a known date. A second group with similar characteristics, the **control group**, is not hit. Both groups are observed **before** and **after** the date, in a panel or in repeated cross-sections.

The method

The effect of the shock is the **change** in the outcome of the treated group minus the **change** in the outcome of the control group.

- In a regression, simple DiD is the TWFE model of the panel chapter with one additional regressor, a dummy equal to 1 for treated units after the shock.

The origin: John Snow and cholera (1855)



Origin: epidemiology — the fight against cholera in London. Cholera was assumed to be transmitted by **air**; John Snow (1855) argued instead that it came from **infected water**.

How to prove it? A *natural experiment*. The whole city drew water from the same point in the Thames, until a reform split the supply: some districts now received water from **downstream** of the sewers (infected), others from **upstream** (assumed uncontaminated).

$\Rightarrow$ *treated vs control*, *before vs after* — the first difference-in-differences.

Roadmap

1. The **canonical** 2 × 2 — potential outcomes, parallel trends, the estimator.
2. **The comparison group** — what makes a control credible.
3. **Dynamics** — the event study.
4. **Diagnostics** — pre-trends, bounds, placebos, composition.
5. **Inference** — serial correlation, and one treated group.

*Recent synthesis: Roth, Sant'Anna, Bilinski & Poe (2023). Textbook treatment: Cunningham, *Causal Inference: The Mixtape*.*

Section 1

The 2 × 2

Potential outcomes and the missing counterfactual

Each unit i has two potential outcomes: $Y_i(1)$ **if treated**, $Y_i(0)$ **if not**. The treatment indicator $D_i \in \{0, 1\}$ selects the one we observe, and the causal effect is the gap between the two:

$$Y_i = D_i\, Y_i(1) + (1 - D_i)\, Y_i(0), \qquad \tau_i = Y_i(1) - Y_i(0).$$

The fundamental problem of causal inference

For any unit we observe **only one** potential outcome. A treated unit reveals $Y_i(1)$; its $Y_i(0)$, what would have happened **without** treatment, is the **missing counterfactual**.

- Individual effects τ_i are not identified; the targets are **averages over groups**.
- Notation: $\mathbb{E}[\,\cdot \mid D_i = 1\,]$ is the average **among treated** units, $\mathbb{E}[\,\cdot \mid D_i = 0\,]$ the average **among untreated** units.

The target: the average effect on the treated

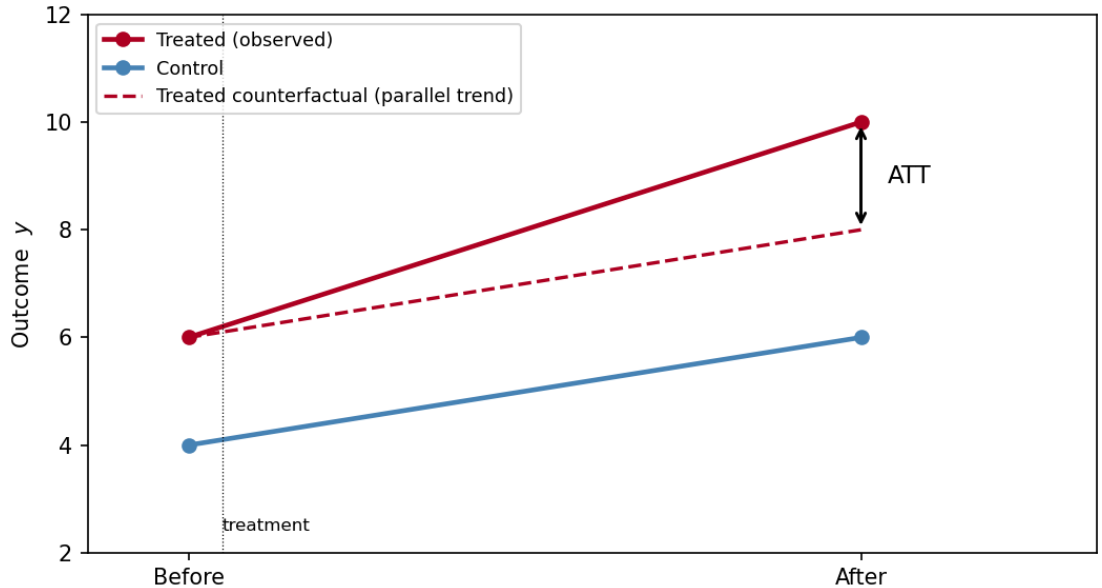
Average treatment effect on the treated (ATT)

$$\text{ATT} = \mathbb{E}[\, Y_i(1) - Y_i(0) \mid D_i = 1 \,].$$

- The effect for the units that **were** treated: did the policy change the outcome of its recipients?
- The average effect on the **untreated**, $\mathbb{E}[\, Y_i(1) - Y_i(0) \mid D_i = 0 \,]$, is a different parameter when effects are heterogeneous; DiD does not identify it.
- **DiD targets the ATT.** The missing piece is the treated group's untreated counterfactual $\mathbb{E}[\, Y_i(0) \mid D_i = 1 \,]$, which **parallel trends** will supply.

The two-by-two: two differences

Two groups, two periods — we now index outcomes by time, $Y_{it}(d)$. DiD builds the treated group's missing counterfactual by assuming it would have moved **like the control**:



$$\widehat{\text{ATT}} = \underbrace{(\bar{y}_{T,\text{post}} - \bar{y}_{T,\text{pre}})}_{\text{treated change}} - \underbrace{(\bar{y}_{C,\text{post}} - \bar{y}_{C,\text{pre}})}_{\text{control change}}$$

Parallel trends: the identifying assumption

Parallel trends

Had the treated group *not* been treated, its outcome would have changed over time by the **same amount** as the control group. Writing $\Delta Y_i(0)$ for that change in the **untreated** outcome:

$$\mathbb{E}[\Delta Y_i(0) \mid D_i = 1] = \mathbb{E}[\Delta Y_i(0) \mid D_i = 0].$$

- **Levels** may differ (fixed effects absorb them); only the **trend** must match.
- **Untestable** (an unobserved counterfactual): defended with pre-period evidence and economic argument, never proved.
- **No anticipation**, the companion assumption: treated units do not respond before the date, so their pre-period outcome is $Y_i(0)$.



The regression form

The 2×2 DiD is one OLS regression:

$$y_{it} = \alpha + \gamma\, D_i + \lambda\, \text{Post}_t + \beta\, (D_i \times \text{Post}_t) + \varepsilon_{it}.$$

- γ — the fixed treated-vs-control gap; λ — the common time shock.
- β — **the DiD coefficient, equal to the ATT** under parallel trends; the interaction $D_i \times \text{Post}_t$ equals d_{it}.
- Most of the time this standard DiD is estimated with a TWFE:

$$y_{it} = f_i + f_t + \beta\, d_{it} + u_{it}.$$

Example: Card and Krueger (1994)

New Jersey raised its minimum wage (1992); neighbouring Pennsylvania did not. Compare fast-food employment across the state line.

FTE employment	Before	After	Change
New Jersey (treated)	20.44	21.03	+0.59
Pennsylvania (control)	23.33	21.17	−2.16
Difference-in-differences			**+2.76**

- Employment in New Jersey **rose** relative to Pennsylvania — not just "did not fall" — against the textbook competitive prediction.
- The cross-border design makes parallel trends plausible: similar labour markets, same macro shocks — the control was chosen for that reason (next section).

Section 2

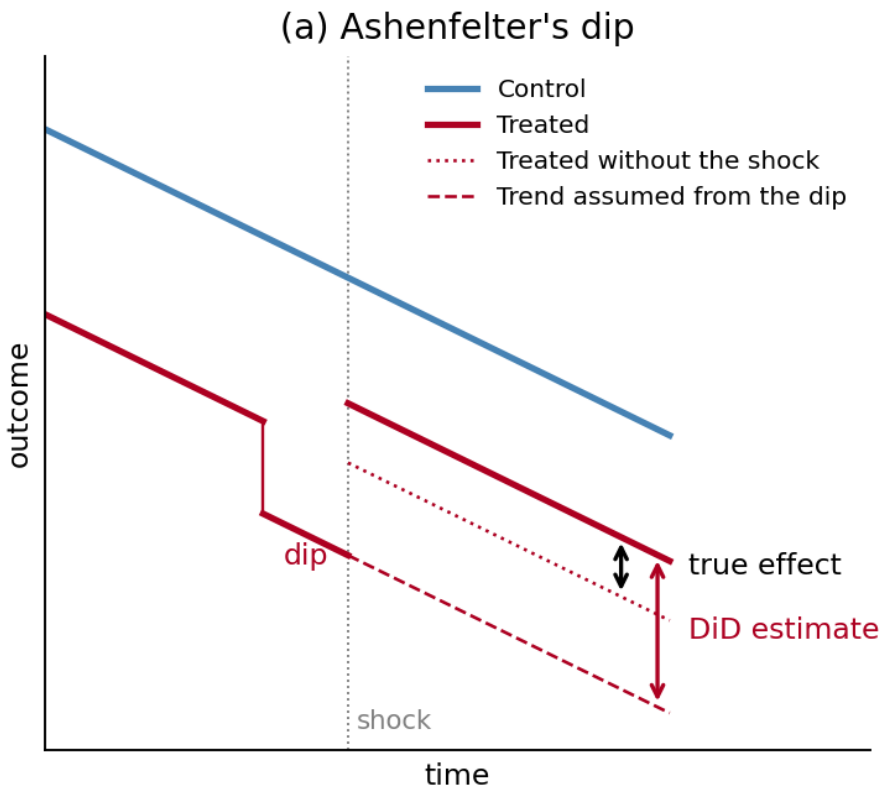
Comparison group

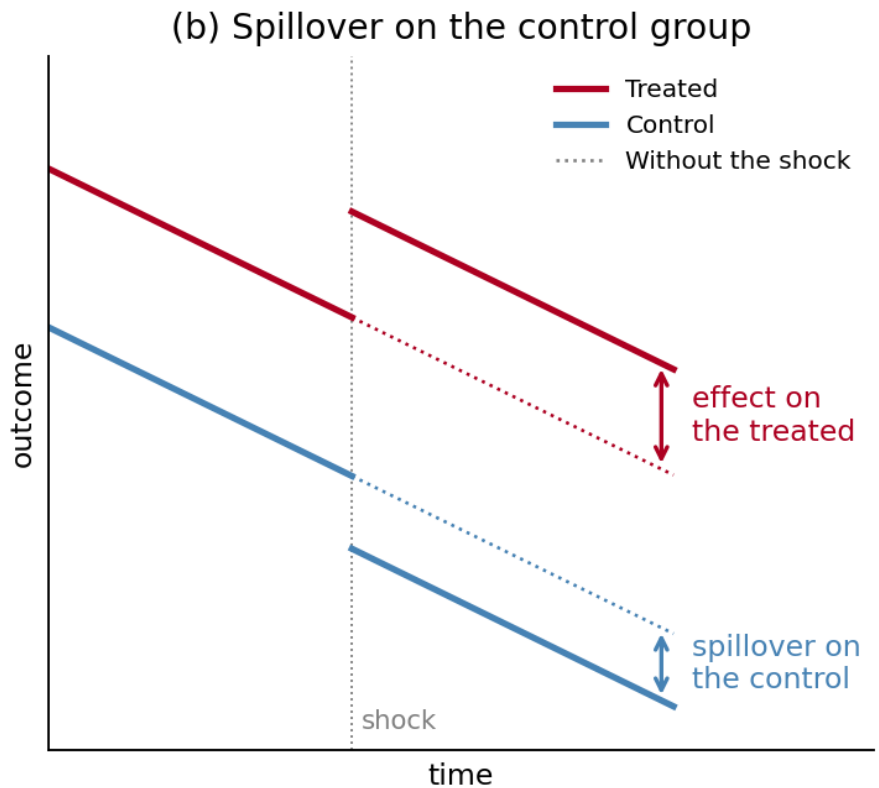
Designing the comparison

Parallel trends is an assumption on the **shocks**: absent the policy, the treated and the comparison group would have received the same ones. The choice of the comparison group is part of the identification strategy and is made on economic grounds before estimation.

1. **Common shocks.** Same industry, markets and geography, so that the time effect f_t is common to both groups, with no shock specific to one group during the estimation window.
2. **No spillovers (SUTVA).** The policy does not change the outcome of the comparison group. Identify the channel of the policy (a price, a quantity, a payment) and verify that it did not move for that group.
3. **No anticipation.** The treated group does not respond before the date. Otherwise the pre-period already contains part of the effect.
4. **No selection on a temporary shock.** Units may differ from the control group in level, but they must not enter treatment because their outcome has just changed temporarily. Example: Ashenfelter's dip (Ashenfelter, 1978). Workers enter a training programme because their earnings have just fallen, and these earnings would have recovered anyway.

Two violations of the design





Section 3

Dynamics

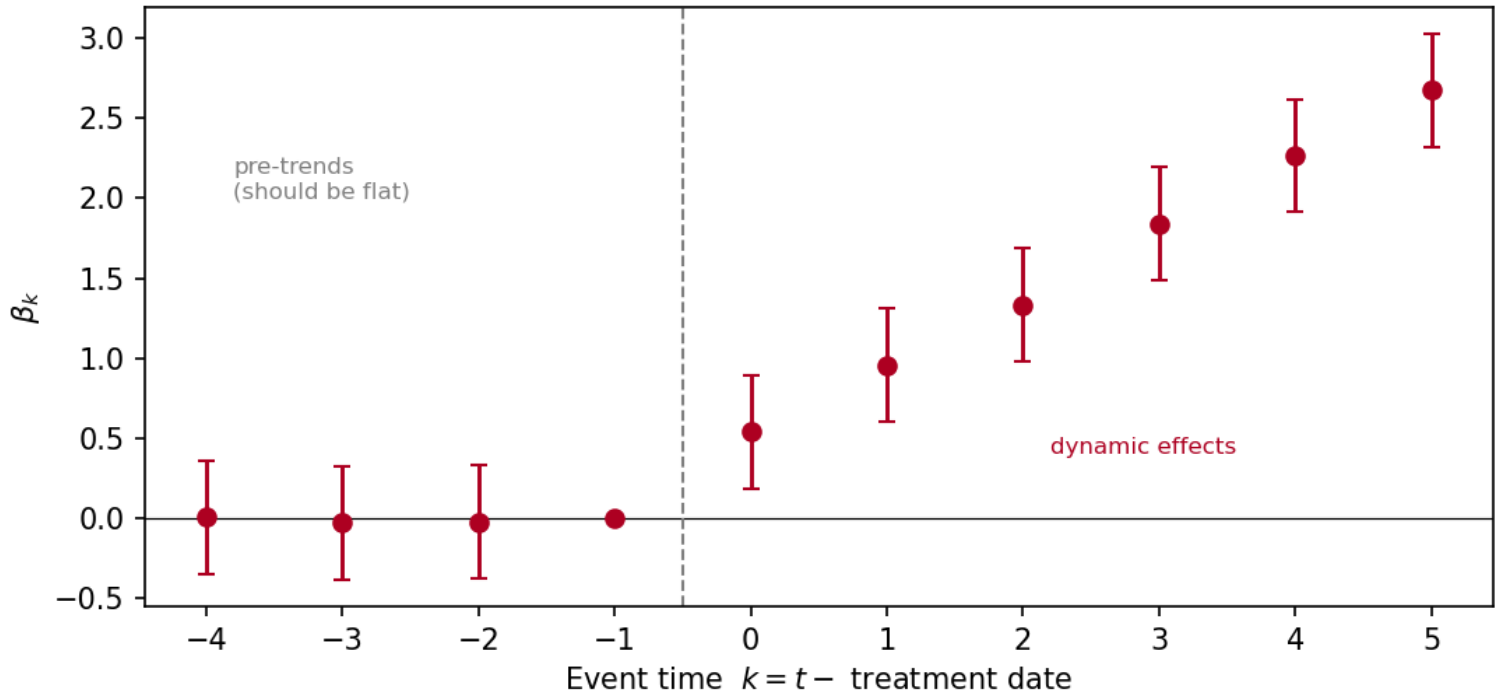
The event-study specification

Replace the single switch by a full set of **leads and lags** around each unit's treatment date:

$$y_{it} = f_i + f_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}\{t - t_i^* = k\} \, D_i + u_{it}.$$

- $k = t - t_i^*$ is **event time** (t_i^* = the treatment date); $k = -1$ is the **omitted baseline**, so each β_k is relative to the period just before treatment.
- β_k for $k < 0$ are **pre-trends**; for $k \geq 0$ they are **dynamic treatment effects**.
- With one treatment date this is simply one coefficient per **calendar year** on the treated dummy.

Reading the event study



- **Flat, near-zero** coefficients before 0 are consistent with parallel trends; the post coefficients trace the effect's **dynamics**.
- A sloping pre-period ($\beta_k \neq 0$ for $k < 0$) is a failure of parallel trends before treatment: the comparison group has to be revised.

A caution on pre-trends

Passing a pre-trend test is reassuring, not decisive

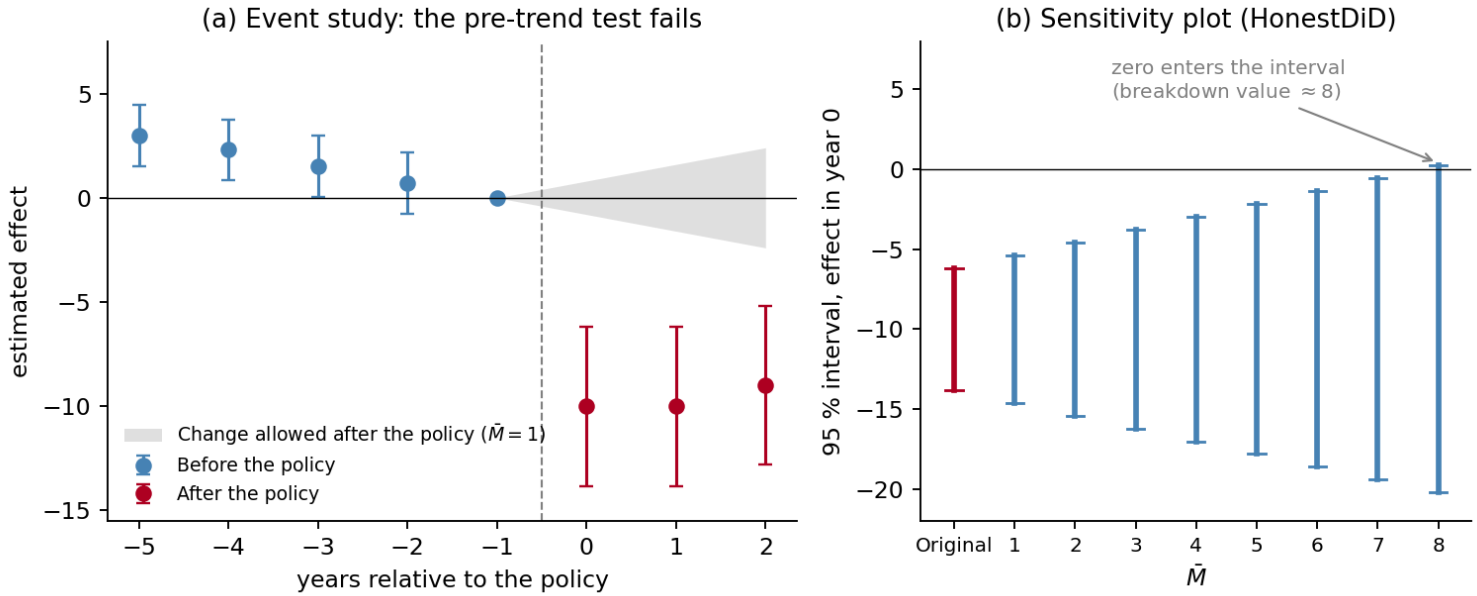
Tests of $\beta_k = 0$ for $k < 0$ often have **low power** — they miss real violations — and conditioning on "passing" can itself distort the estimate (Roth, 2022).

- Treat them as **evidence**, alongside an economic case for why the trends should align.
- **Plot** the coefficients; never rely on a single p-value (on event-study design and visualization, see Freyaldenhoven et al. 2021).

Section 4

Diagnostics

Sensitivity analysis: Rambachan and Roth (2023)



- **(a)** Pre-trend test fails. The largest change from one year to the next is 0.8. If the gap kept changing that much after the policy (grey band, $\bar{M} = 1$), the estimated effects would still lie far outside the band.
- **(b)** The bars stay below zero up to $\bar{M} \approx 8$. Despite the failed pre-trend test, the conclusion that the policy lowered the outcome holds, because the violation of parallel trends is small compared with the effect. A bar crossing zero already at $\bar{M} < 1$ would mean that changes as large as those observed before the policy could explain the whole effect.

Placebo tests

A design that identifies an effect where there is one should find **nothing** where there is none:

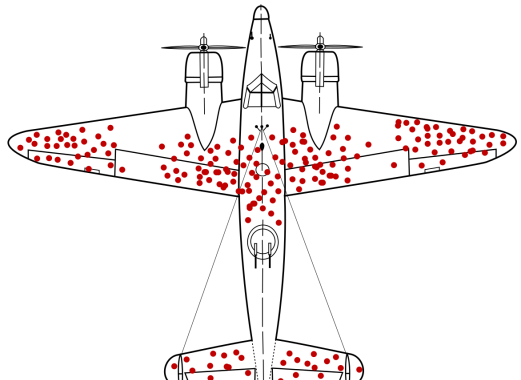
- **Placebo in time** — apply the design to a date at which nothing happened, inside the pre-period.
- **Placebo group** — apply the design to a group that could not have been affected.
- **Placebo outcome** — an outcome the policy cannot reach through the claimed channel.

A placebo in time is a 2×2 on pre-treatment years, a linear combination of the pre-period coefficients: a rejection is a failure of parallel trends at that date. Placebo groups and outcomes test the channel: a rejection means some other shock separates the groups. Both belong in the paper.

Composition: the balanced-panel check

Units enter and leave panels — and **leaving can be an effect** of the policy (firms close, farms exit, workers move).

- In the full sample the post-period mean is computed on **survivors**; if the leavers were the losers, the effect appears to fade faster than it does for any given unit.
- **Check:** re-estimate on units present in every year of a window around the policy (a *balanced* panel).
- Report **both**. They answer different questions: the balanced panel gives the effect on the units that stay, the full panel the effect on the observed population, composition included. The direction of the gap depends on who leaves.



Section 5

Inference

Autocorrelation and standard errors

A DiD panel follows the same units for many years. Within a unit, two variables change little from one year to the next:

- the outcome y_{it}, which is autocorrelated (panel chapter);
- the treatment d_{it}, which equals 1 in every year after the policy.

The years of one unit are therefore not independent observations, and standard errors that treat them as independent are too small.

- **Solution** (Bertrand et al., 2004). Cluster the standard errors at the level at which the treatment is assigned. If each firm is treated or not, cluster by firm: all its years form one cluster, and the correlation between them is allowed.
- When the policy is assigned to whole groups of units (all the firms of a region), cluster by that group, because its units share the treatment and its shocks.
- Report G, the number of clusters. The estimator requires a large G.

One treated group

Many DiDs have a single treated group: a sector, a state, a country. Two problems follow for the standard errors.

- Clustering at the level of treatment assignment gives $G = 2$ clusters, the treated group and the comparison group. The cluster-robust estimator requires a large G and cannot be used.
- Clustering by unit ignores the shock that hits the whole treated group in a given year (a price cycle, an embargo). With one treated group this shock does not average out, and the standard errors are too small.

1. **Placebo in time.** Estimate the same DiD at dates before the policy, when the true effect is zero. The estimates obtained show how large an effect the shocks alone produce.
2. **Permutation test.** When several untreated groups are available, assign the treatment to each of them in turn and re-estimate. If the true estimate is larger than almost all of these placebo estimates, the shocks alone are unlikely to explain it (Conley and Taber, 2011; Ferman and Pinto, 2019).

Section 6

Conclusion

Conclusion

1. The objective of a DiD analysis is to estimate the average effect of a policy on the treated group (ATT). It is identified under parallel trends and no anticipation.
2. The comparison group is chosen on economic grounds, before estimation: common shocks, no spillovers, no anticipation, no Ashenfelter's dip.
3. The 2×2 gives one average effect. The event study gives the dynamic effect.
4. The result is then checked with the pre-period coefficients, a sensitivity analysis, placebo tests and a balanced panel.

Next chapter: units treated at different dates.

Section 7

Applied

Build the treatment variables

Six lines create the treatment variables, the region × year cell, the sample and the clustering.

```
df["lait"] = (df["groupe"] == "lait").astype(int)                    # 1
df["post"] = (df["annee"] >= 2015).astype(int)                       # 2
df["d"]    = df["lait"] * df["post"]                                 # 3
df["reg_an"] = df["region"].astype(str) + "_" \
             + df["annee"].astype(str)                               # 4
base = df[df["groupe"].isin(["lait", "caprins"])]                    # 5
cl = {"CRV1": "id"}                                                  # 6
```

- **1**, **2** — D_i, 1 for a dairy farm, and Post_t, 1 from 2015 on; `.astype(int)` turns `True`/`False` into 1/0.
- **3** — $d_{it} = D_i \times \text{Post}_t$: the regressor whose coefficient is the ATT.
- **4** — a text label such as `"22_2015"`, one per region and year; its fixed effect compares dairy and goat farms of the **same region the same year**.
- **5** — keeps the treated group and its control; `.isin` tests membership in a list.
- **6** — standard errors clustered by farm, passed to every regression as `vcov=cl`.

The 2 × 2: by hand, then in regression

Five lines reproduce the table of four means and three regression versions, equal only in a balanced panel.

```
w = base[base["annee"].between(2012, 2016)]                          # 1
tab = w.groupby(["groupe","post"])["res_uta"].mean().unstack()       # 2
m1 = pf.feols("res_uta ~ lait + post + lait:post", w, vcov=cl)       # 3
m2 = pf.feols("res_uta ~ d | id + annee", w, vcov=cl)                # 4
m3 = pf.feols("res_uta ~ d | id + reg_an", w, vcov=cl)               # 5
```

- **1** — three years before, two years after.
- **2** — the mean outcome by group and period, `unstack()` putting the two periods side by side: the 2 × 2 table. The DiD is the difference of the two row changes.
- **3** — the same number as a regression: `lait:post` is the interaction, its coefficient is β; `lait` and `post` alone are γ and λ.
- **4** — with farm and year fixed effects: each farm compared to itself.
- **5** — with region × year effects instead of year effects: neighbours only.

The event study

Three lines estimate one coefficient per year and read them.

```
sc = base[~base["annee"].isin(range(2007, 2012))]                    # 1
es = pf.feols("res_uta ~ i(annee, lait, ref=2014) | id + reg_an",
              sc, vcov=cl)                                           # 2
es.coef()                                                            # 3
```

- **1** — drops the years 2007 to 2011, the dairy cycle excluded ex ante; `~` negates the condition.
- **2** — `i(annee, lait, ref=2014)` builds one dummy per year, **multiplied by** the treated indicator, and leaves 2014 out as the reference. The coefficient of each dummy is β_k: the dairy-minus-goat gap in that year, relative to 2014.
- **3** — the coefficients, named `annee::2015:lait` and so on. The years before 2015 are the pre-trend check, the years after trace the effect. The four-digit year can be read out of each name with a regular expression.

Pre-trend test, placebo date, balanced panel

Three groups of lines produce the checks of the chapter.

```
b = es.coef()[pre]; V = es._vcov[np.ix_(idx, idx)]                    # 1
W = b @ np.linalg.solve(V, b)                                         # 2
p = 1 - stats.chi2.cdf(W, len(b))                                     # 3
pl = base[base["annee"].between(2000, 2006)].copy()                   # 4
pl["d_pl"] = pl["lait"] * (pl["annee"] >= 2004).astype(int)           # 5
bal = base[base["id"].isin(ids_bal)
           & base["annee"].between(2012, 2018)]                       # 6
```

- **1** to **3** — the joint test that all pre-2015 coefficients are zero. `pre` lists their names, `idx` their positions; `b` is the vector of estimates, `V` their covariance block (`np.ix_` selects it); `W = b'V^{-1}b` is the Wald statistic, χ^2 under H_0 with as many degrees of freedom as coefficients.
- **4**, **5** — the placebo: pre-period only, a fake treatment date in 2004; the coefficient of `d_pl` should be near zero.
- **6** — the balanced panel: `ids_bal` lists the farms present every year from 2012 to 2018; the sample keeps them over those years, so the set of farms observed no longer changes.

References I

Ashenfelter, Orley. 1978. "Estimating the Effect of Training Programs on Earnings." *The Review of Economics and Statistics* 60 (1): 47–57. https://doi.org/10.2307/1924332.

Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *The Quarterly Journal of Economics* 119 (1): 249–75. https://doi.org/10.1162/003355304772839588.

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84 (4): 772–93.

Conley, Timothy G., and Christopher R. Taber. 2011. "Inference with 'Difference in Differences' with a Small Number of Policy Changes." *Review of Economics and Statistics* 93 (1): 113–25. https://doi.org/10.1162/REST_a_00049.

Cunningham, Scott. 2021. *Causal Inference: The Mixtape*. Yale University Press.

Ferman, Bruno, and Cristine Pinto. 2019. "Inference in Differences-in-Differences with Few Treated Groups and Heteroskedasticity." *Review of Economics and Statistics* 101 (3): 452–67. https://doi.org/10.1162/rest_a_00759.

Freyaldenhoven, Simon, Christian Hansen, Jorge Pérez Pérez, and Jesse M. Shapiro. 2021. *Visualization, Identification, and Estimation in the Linear Panel Event-Study Design*. NBER Working Paper No. 29170. National Bureau of Economic Research.

Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225 (2): 254–77. https://doi.org/10.1016/j.jeconom.2021.03.014.

Rambachan, Ashesh, and Jonathan Roth. 2023. "A More Credible Approach to Parallel Trends." *Review of Economic Studies* 90 (5): 2555–91. https://doi.org/10.1093/restud/rdad018.

Roth, Jonathan. 2022. "Pretest with Caution: Event-Study Estimates After Testing for Parallel Trends." *American Economic Review: Insights* 4 (3): 305–22. https://doi.org/10.1257/aeri.20210236.

References II

Roth, Jonathan, Pedro H. C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." *Journal of Econometrics* 235 (2): 2218–44. https://doi.org/10.1016/j.jeconom.2023.03.008.